

# Class #19 - Properties of Determinants

1. Determinants of row equivalent matrices:

$\det I_{n \times n} = \underline{\hspace{2cm}}$  (triangular matrix)

$E_1$  (row - interchange):  $\underline{\hspace{2cm}}$  (at  $2 \times 2$  stage)

$E_2$  (scaling):  $\underline{\hspace{2cm}}$

$E_3$  (replacement):  $\underline{\hspace{2cm}}$  (because of zeros)

2. Let  $E$  be an elementary matrix (matrix obtained after single row operation on identity matrix) and  $A$  be a square matrix. Then,  $\det EA = \underline{\hspace{2cm}}$  where  $\det E = 1, -1,$  or  $c$ .

Sketch of the proof: can be shown by induction.

Check by direct calculation for  $2 \times 2$ .

For  $n \times n$ ,  $\det EA =$  (expansion on unchanged row which exists, say row  $i$ )

$$= a_{i1}(-1)^{i+1} \det B_{i1} + \cdots + a_{in}(-1)^{i+n} \det B_{in}$$

$$= \alpha a_{i1}(-1)^{i+1} \det A_{i1} + \cdots + \alpha a_{in}(-1)^{i+n} \det A_{in}$$

$$= \alpha \cdot \det A = \det E \det A$$

3. Example: How are  $\det A$  and  $\det B$  are related?

$$A = \begin{bmatrix} 3 & 5 & 6 \\ 1 & \pi & e \\ 0 & 0 & 2 \end{bmatrix} \quad B = \begin{bmatrix} \frac{3}{2} & \frac{5}{2} & 3 \\ 1 & \pi & e \\ 0 & 0 & 2 \end{bmatrix}$$

4. Example: If  $B$  is obtained from  $A$  by 3 row interchanges, then  $\det B = \underline{\hspace{2cm}}$   $\det A$ .

5. Find the determinant of  $A$  by row reduction.

$$A = \begin{bmatrix} 1 & -1 & -3 & 0 \\ 0 & 1 & 5 & 4 \\ -1 & 2 & 8 & 5 \\ 3 & -1 & -2 & 3 \end{bmatrix}$$

$\det A = \underline{\hspace{2cm}} \cdot (\underline{\hspace{2cm}}) \neq 0$  if  $A \underline{\hspace{2cm}} I$ , so  $A$  is  
 $\underline{\hspace{2cm}}$ .

6. Determinant of a product of  $n \times n$  matrices:

$$\det AB = \underline{\hspace{2cm}}$$

Proof:

(a) If  $A$  is invertible:

$$E_k \cdots E_2 E_1 A = I, \text{ so } A = \underline{\hspace{2cm}}, \text{ and } \det A = \underline{\hspace{2cm}}.$$

$$\text{Then, } \det AB = \det \underline{\hspace{2cm}} B = \underline{\hspace{2cm}} = \det A \det B.$$

(b) If  $A$  is not invertible:

$$\text{Then } AB \text{ is not } \underline{\hspace{2cm}}. \text{ (Why?)}$$

$$\text{So, } \det AB = 0 = \det A \det B.$$

(c) If  $B$  is not invertible: